

Power Series

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1.5 Power Series

General Properties of Power Series

It is well known that $(1-x)^{-1} = \sum_{k=0}^{\infty} x^k$ when $|x| < 1$. Similar expansions can be obtained from Taylor's formula,

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n-1)}(0)}{(n-1)!}x^{n-1} + R_n(x),$$

for those values of x for which the remainder $R_n(x)$ tends to 0 as $n \rightarrow \infty$. For instance, if $f(x) = e^x$, then we have

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^{n-1}}{(n-1)!} + R_n(x), \quad R_n(x) = e^{\xi} \frac{x^n}{n!},$$

where $|\xi| \leq |x|$. This can be written as

$$\sum_{k=0}^{n-1} \frac{x^k}{k!} = e^x - R_n(x),$$

and thus we have found an expression for the n^{th} partial sum of the series

$$\sum_{k=0}^{\infty} \frac{x^k}{k!}. \quad (11)$$

Since $x^n/n! \rightarrow 0$ as $n \rightarrow \infty$ for each fixed x , the limit of $R_n(x)$ is 0 as $n \rightarrow \infty$. This means that the series (11) is convergent to e^x for all values of x . By replacing x with $|x|$, we see that the series is, in fact, absolutely convergent.

A (real) *power series* is a series of the form

$$\sum_{k=0}^{\infty} a_k x^k \quad (12)$$

where a_k are real numbers and x is a real variable. The series in (11) is an example of such series. The other elementary functions can be treated analogously but we postpone this discussion until we have studied power series in general.

Lemma 1. Assume that the terms of the power series (12) form a bounded sequence of numbers for some $x = x_0 \neq 0$. Then the series is absolutely convergent when $|x| < |x_0|$.

Proof. Since $|a_k x_0^k| \leq C$ for all k , we have that

$$|a_k x^k| \leq C \left| \frac{x}{x_0} \right|^k, \quad k = 0, 1, 2, \dots$$

By comparison with the geometric series, the power series is absolutely convergent when $|x/x_0| < 1$. ■

Definition. The number

$$\sup\{|x|; \text{ the sequence } (a_k x^k)_0^\infty \text{ is bounded}\}$$

is called the *radius of convergence* of the power series (12).

We shall denote the radius of convergence by R . The reason for using the term “radius” is that we can allow x to take on complex values and then the equation $|x| = R$ represents a circle in the complex plane. We shall return to this in Section 1.6¹.

The radius of convergence can be characterised in the following way.

Theorem 5.1. If the power series (12) has radius of convergence R , then

- (1) the series is absolutely convergent when $|x| < R$,
- (2) the series is divergent when $|x| > R$.

Proof. (1) Assume that $R > 0$ and that $|x| < R$. From the definition of R we can find an x_0 such that $|x| < |x_0| < R$ and for which the sequence $(a_k x_0^k)_0^\infty$ is bounded. Hence, by the lemma, the power series is absolutely convergent at x .

(2) If $|x| > R$, then the terms of the series are unbounded. Therefore they do not tend to zero and hence the power series is divergent. ■

Remark. It follows from the theorem that the radius of convergence also is the supremum of those $|x|$ for which $\sum_{k=0}^\infty a_k x^k$ is convergent.

If $R = 0$, then the power series is convergent only when $x = 0$. If $R = \infty$, then it is absolutely convergent for all x . If $0 < R < \infty$, then the following example shows that nothing can be said about the convergence when $|x| = R$.

Example 5.1. Consider the three power series

$$(a) \sum_{k=1}^{\infty} x^k, \quad (b) \sum_{k=1}^{\infty} \frac{x^k}{k}, \quad (c) \sum_{k=1}^{\infty} \frac{x^k}{k^2}.$$

Their radii of convergence are all equal to 1 because the terms of the series are bounded when $|x| < 1$ and unbounded when $|x| > 1$. For instance, regarding the series (b), we have that

$$\left| \frac{x^k}{k} \right| \leq 1 \quad \text{for all } k \quad \text{if} \quad |x| < 1$$

whereas

$$\left| \frac{x^k}{k} \right| \rightarrow \infty \quad \text{as } k \rightarrow \infty \quad \text{if} \quad |x| > 1.$$

The series (a) is divergent when $x = \pm 1$ (because the terms do not tend to zero). The series (b) is divergent when $x = 1$ and conditionally convergent when $x = -1$. Finally, the series (c) is absolutely convergent when $|x| = 1$.

In simple cases, the radius of convergence can be determined directly from the definition as in Example 5.1. Sometimes, though, the following formula is convenient.

Theorem 5.2 (D’Alembert’s ratio test). If $a_k \neq 0$ for all k and the limit

$$L = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right|$$

exists (finitely or infinitely), then the radius of convergence of the power series $\sum_{k=0}^\infty a_k x^k$ is $R = 1/L$.

¹Not contained in this excerpt.

Proof. This follows immediately from Theorem 2.3². In fact, when $x \neq 0$,

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}x^{k+1}}{a_kx^k} \right| = |x| \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = |x|L.$$

Hence the series is absolutely convergent when $|x|L < 1$, i.e. when $|x| < 1/L$, and divergent when $|x|L > 1$, i.e. when $|x| > 1/L$. If $L = 0$, then the series is convergent for all x , i.e. $R = \infty$, and if $L = \infty$, then the series is convergent only when $x = 0$, i.e. $R = 0$. ■

Example 5.2. Determine the radius of convergence of the series

$$\sum_{k=0}^{\infty} \binom{2k}{k} x^k.$$

Solution: Since

$$\binom{2k}{k} = \frac{(2k)!}{(k!)^2},$$

we have

$$\frac{a_{k+1}}{a_k} = \frac{(2k+2)!}{((k+1)!)^2} \cdot \frac{(k!)^2}{(2k)!} = \frac{(2k+2)(2k+1)}{(k+1)^2} = 2 \cdot \frac{2k+1}{k+1} \rightarrow 4 \quad \text{as } k \rightarrow \infty.$$

Hence $R = 1/4$.

Example 5.3. Determine the radius of convergence of the power series

$$\sum_{k=0}^{\infty} \frac{x^{2k}}{3^k + 1}.$$

Solution: This series contains only even powers of x , i.e. every other coefficient a_k is zero. D'Alembert's ratio test can therefore not be applied directly. This difficulty can be circumvented if we think of x^2 as the variable and study the series

$$\sum_{k=0}^{\infty} \frac{1}{3^k + 1} (x^2)^k.$$

We find that the radius of convergence of this series is 3 because

$$\frac{3^k + 1}{3^{k+1} + 1} = \frac{1 + 3^{-k}}{3 + 3^{-k}} \rightarrow \frac{1}{3} \quad \text{as } k \rightarrow \infty.$$

The series is therefore absolutely convergent when $x^2 < 3$ and divergent when $x^2 > 3$. Consequently, its radius of convergence is $R = \sqrt{3}$.

Example 5.4. Determine the radius of convergence of the power series

$$\sum_{k=0}^{\infty} x^{k^2}.$$

Solution: Here the coefficients are zero except when their indices are perfect squares. Therefore, d'Alembert's ratio test cannot be used, and the trick used in the previous example does not work. However, using the definition of R works fine here. We see immediately that the sequence $(x^{k^2})_1^\infty$ is bounded when $|x| < 1$ and tends to ∞ when $|x| > 1$. Hence $R = 1$.

²The theorem, not contained in this excerpt, states the ratio test for positive series.

Differentiation and Integration of Power Series

The sum $s(x)$ of a convergent power series $\sum_0^\infty a_k x^k$ is of course a function of x . Now we shall see that this function is differentiable and that its derivative can be obtained by differentiating each term of the series. We prepare for the proof of this by examining the differentiability at the origin.

Lemma 2. Assume that the power series $\sum_0^\infty a_k x^k$ has a positive radius of convergence R . Then its sum $s(x)$ is differentiable at the origin and $s'(0) = a_1$.

Proof. If $|h| < R$, then

$$s(h) = a_0 + a_1 h + h^2 B(h)$$

where

$$B(h) = a_2 + a_3 h + \cdots = \sum_{k=0}^\infty a_{k+2} h^k.$$

Choose r so that $0 < r < R$. Since the power series $B(h)$ is absolutely convergent when $h = r$, we have that $\sum_0^\infty |a_{k+2}| r^k = M < \infty$. Hence, by the triangle inequality,

$$|s(h) - a_0 - a_1 h| = |h^2 B(h)| \leq M |h|^2 \quad \text{when } |h| \leq r.$$

Since $s(0) = a_0$, this means that

$$\left| \frac{s(h) - s(0)}{h} - a_1 \right| \leq M |h| \quad \text{when } |h| \leq r,$$

and now we see that $s(x)$ is differentiable at the origin with derivative a_1 . (Let $h \rightarrow 0$.) ■

Theorem 5.3. Assume that the power series $\sum_0^\infty a_k x^k$ has radius of convergence $R > 0$. Then its sum $s(x)$ is a continuous function of x when $|x| < R$. Furthermore, $s(x)$ is differentiable when $|x| < R$ and its derivative is

$$s'(x) = \sum_{k=1}^\infty k a_k x^{k-1}, \quad (13)$$

which is a power series with the same radius of convergence R .

Proof. The continuity follows from the differentiability.

Fix an x such that $|x| < R$ and choose h so that $|x| + |h| < R$. Then

$$\sum_{k=0}^\infty |a_k| (|x| + |h|)^k < \infty. \quad (14)$$

Now consider the following table.

a_0	0	0	$\cdots$	0	0	$\cdots$
$a_1 x$	$a_1 h$	0	$\cdots$	0	0	$\cdots$
$a_2 x^2$	$2a_2 x h$	$a_2 h^2$	$\cdots$	0	0	$\cdots$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$	
$a_n x^n$	$na_n x^{n-1} h$	$\binom{n}{2} a_n x^{n-2} h^2$	$\cdots$	$a_n h^n$	0	$\cdots$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$	

The sum of the elements in row $n + 1$ is, according to the binomial theorem, equal to $a_n(x + h)^n$. Hence the sum of all the row sums is

$$\sum_{n=0}^{\infty} a_n(x + h)^n = s(x + h).$$

If we instead form the sum of all the column sums, then we get an expression

$$A_0(x) + A_1(x)h + A_2(x)h^2 + \cdots$$

where $A_0(x) = s(x)$,

$$A_1(x) = a_1 + 2a_2x + \cdots + na_nx^{n-1} + \cdots = \sum_{k=1}^{\infty} ka_kx^{k-1},$$

and so forth. The double series formed by all the terms in the table is absolutely convergent, for if we replace each term with its absolute value and sum row-wise in the same way as above, then we get the series (14). (The sum of a positive series is independent of the order of summation.) From the absolute convergence it follows that row-wise and column-wise summations in the table give the same result. Hence

$$s(x + h) = A_0(x) + A_1(x)h + A_2(x)h^2 + \cdots. \quad (15)$$

We remind that x is fixed and that the variable is h . In (15), we have written $s(x + h)$ as the sum of a power series in h . This series is convergent when $|h| < R - |x|$. Lemma 2 now yields that $s(x + h)$ is differentiable at $h = 0$ with derivative $A_1(x)$. This shows that $s(x)$ is differentiable at x and that Formula (13) is valid.

It remains to show that the radius of convergence R' of the differentiated series (13) is equal to R . It is plain that $R' \geq R$ because the series $A_1(x)$ is convergent when $|x| < R$. The reverse inequality $R \geq R'$ follows from the comparison theorem because the inequality

$$|a_kx^k| \leq |x| |ka_kx^{k-1}|, \quad k \geq 1,$$

shows that the given power series is convergent if the differentiated power series is (absolutely) convergent. (This also shows that if $R = 0$, then $R' = 0$.) The proof of Theorem 5.3 is complete. ■

Corollary. If the power series $s(x) = \sum_0^{\infty} a_kx^k$ has a positive radius of convergence R , then

$$S(x) = a_0x + \frac{a_1x^2}{2} + \frac{a_2x^3}{3} + \cdots = \sum_{k=0}^{\infty} a_k \frac{x^{k+1}}{k+1} \quad (16)$$

has the same radius of convergence and $S'(x) = s(x)$ when $|x| < R$.

Proof. Termwise differentiation of the power series (16), in agreement with Theorem 5.3, gives $\sum_0^{\infty} a_kx^k$. ■

Example 5.5. Compute the sum of the series $\sum_0^\infty kx^k$.

Solution: We start from the geometric series

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \quad |x| < 1.$$

Differentiation according to Theorem 5.3 gives

$$\sum_{k=1}^{\infty} kx^{k-1} = (1-x)^{-2}, \quad |x| < 1. \quad (17)$$

From this it follows that $\sum_1^\infty kx^k = x(1-x)^{-2}$ with convergence when $|x| < 1$.

Remark. Earlier, in Example 3.3³, we derived Formula (17) in a more complicated manner.

Consider anew $s(x) = \sum_0^\infty a_k x^k$, $|x| < R$. Since $s'(x)$ is given by the power series (13) with radius of convergence R , we can apply Theorem 5.3 to this series. Then we find that $s''(x)$ exists and is equal to

$$\sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}, \quad |x| < R.$$

This can of course be repeated as many times as we want. Hence the function $s(x)$ is infinitely differentiable when $|x| < R$ and

$$\begin{aligned} s^{(n)}(x) &= \sum_{k=n}^{\infty} k(k-1)\cdots(k-n+1)a_k x^{k-n} \\ &= n!a_n + (n+1)!a_{n+1}x + \frac{(n+2)!}{2!}a_{n+2}x^2 + \cdots \end{aligned}$$

In particular, $s^{(n)}(0) = n!a_n$, i.e. $a_n = s^{(n)}(0)/n!$. This shows that the sum $s(x)$ uniquely determines the coefficients a_n : if $s(x) = \sum_0^\infty a_n x^n = \sum_0^\infty b_n x^n$ for all small x , then $a_n = b_n$ for all n .

Expansion in Power Series

We can expand an infinitely differentiable function f in a power series by increasing the number of terms in Taylor's formula indefinitely:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k. \quad (18)$$

In order that the series be convergent and have $f(x)$ as its sum, it is, however, required that the remainder term $R_n(x)$ in Taylor's formula tend to zero as $n \rightarrow \infty$. This is not always the case. As we saw in the beginning of this section, it is true for the exponential function.

We call (18) a power series expansion of the function f . Often, the simplest way to infer convergence of the series is to use d'Alembert's ratio test and the theorem on differentiation of power series. We illustrate this using e^x once again.

³Not contained in this excerpt.

The power series

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

has radius of convergence

$$R = \lim_{k \rightarrow \infty} \frac{(k+1)!}{k!} = \lim_{k \rightarrow \infty} (k+1) = \infty.$$

To compute the sum $s(x)$ of the series, we differentiate the series and get

$$s'(x) = \sum_{k=1}^{\infty} \frac{kx^{k-1}}{k!} = \sum_{k=1}^{\infty} \frac{x^{k-1}}{(k-1)!} = \sum_{p=0}^{\infty} \frac{x^p}{p!} = s(x).$$

Therefore, s is a solution of the differential equation $u' = u$. As is well known, the general solution of this equation is $u(x) = Ce^x$ where C is a constant. Using the power series expansion, we see that $s(0) = 1$, and hence $s(x) = e^x$.

We shall now treat the other elementary functions using the same trick.

Theorem 5.4. The following statements hold.

$$\begin{aligned} \text{(a)} \quad e^x &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots = \sum_{k=0}^{\infty} \frac{1}{k!} x^k, \quad x \in \mathbf{R}. \\ \text{(b)} \quad \cos x &= 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots = \sum_{p=0}^{\infty} \frac{(-1)^p}{(2p)!} x^{2p}, \quad x \in \mathbf{R}, \\ \sin x &= x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots = \sum_{p=0}^{\infty} \frac{(-1)^p}{(2p+1)!} x^{2p+1}, \quad x \in \mathbf{R}. \\ \text{(c)} \quad (1+x)^a &= 1 + ax + \frac{a(a-1)}{2} x^2 + \cdots = \sum_{k=0}^{\infty} \binom{a}{k} x^k, \quad |x| < 1, \end{aligned}$$

where a is an arbitrary real number,

$$\begin{aligned} \binom{a}{0} &= 1 \quad \text{and} \quad \binom{a}{k} = \frac{a(a-1) \cdots (a-k+1)}{k!}, \quad k = 1, 2, \dots \\ \text{(d)} \quad \ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{k+1} x^{k+1}, \quad |x| < 1. \\ \text{(e)} \quad \arctan x &= x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots = \sum_{p=0}^{\infty} \frac{(-1)^p}{2p+1} x^{2p+1}, \quad |x| < 1. \\ \text{(f)} \quad \arcsin x &= x + \frac{x^3}{6} + \frac{3x^5}{40} + \cdots = \sum_{p=0}^{\infty} \frac{(2p-1)!!}{(2p)!!(2p+1)} x^{2p+1}, \quad |x| < 1, \end{aligned}$$

where the semi-factorial $n!!$ is defined by

$$(2p-1)!! = (2p-1)(2p-3) \cdots 3 \cdot 1, \quad (2p)!! = (2p)(2p-2) \cdots 4 \cdot 2$$

with the conventions $(-1)!! = 0!! = 1$.

Proof. That the radii of convergence are the ones stated in the theorem follows from d'Alembert's ratio test. We leave it to the reader to check this out. In the cases (b), (e) and (f), factor out x if necessary and regard the series as a power series in x^2 .

(b) Set

$$c(x) = \sum_{p=0}^{\infty} \frac{(-1)^p}{(2p)!} x^{2p}, \quad s(x) = \sum_{p=0}^{\infty} \frac{(-1)^p}{(2p+1)!} x^{2p+1}.$$

We have that $c(0) = 1$ and $s(0) = 0$. Differentiation yields

$$c'(x) = \sum_{p=1}^{\infty} \frac{(-1)^p}{(2p-1)!} x^{2p-1} = \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{(2k+1)!} x^{2k+1} = -s(x)$$

(we have replaced the index p with $k+1$) and, analogously,

$$s'(x) = c(x).$$

Hence $c'' = -s' = -c$ and $s'' = c' = -s$, which means that the functions s and c both are solutions of the differential equation $u'' + u = 0$. The general solution of this equation is

$$u(x) = A \cos x + B \sin x$$

where A and B are constants. The initial conditions $c(0) = 1$, $c'(0) = 0$, $s(0) = 0$ and $s'(0) = 1$ yield that $c(x) = \cos x$ and $s(x) = \sin x$.

(c) We first observe that

$$\binom{a}{k} = \frac{a(a-1) \cdots (a-k+2)(a-k+1)}{k!} = \binom{a}{k-1} \frac{a-k+1}{k}$$

and get that

$$k \binom{a}{k} = a \binom{a}{k-1} - (k-1) \binom{a}{k-1}.$$

If we set $s(x) = \sum_{k=0}^{\infty} \binom{a}{k} x^k$, then

$$\begin{aligned} s'(x) &= \sum_{k=1}^{\infty} \binom{a}{k} k x^{k-1} = \sum_{k=1}^{\infty} a \binom{a}{k-1} x^{k-1} - \sum_{k=1}^{\infty} (k-1) \binom{a}{k-1} x^{k-1} \\ &= a \sum_{p=0}^{\infty} \binom{a}{p} x^p - x \sum_{p=0}^{\infty} p \binom{a}{p} x^{p-1} = as(x) - xs'(x). \end{aligned}$$

Hence the function s is a solution of the differential equation $(1+x)u' = au$. Since this equation has the general solution $u(x) = C(1+x)^a$ and $s(0) = 1$, it follows that $s(x) = (1+x)^a$.

(d) With

$$s(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k+1} x^{k+1}$$

we have

$$s'(x) = \sum_{k=0}^{\infty} (-1)^k x^k = \sum_{k=0}^{\infty} (-x)^k = \frac{1}{1-(-x)} = \frac{1}{1+x}.$$

Therefore $s(x) = \ln(1+x) + C$, where $C = 0$ because $s(0) = 0$.

(e) Set

$$s(x) = \sum_{p=0}^{\infty} \frac{(-1)^p}{2p+1} x^{2p+1}.$$

Calculations similar to those in (d) show that

$$s'(x) = \frac{1}{1+x^2},$$

and so $s(x) = \arctan x$ because $s(0) = 0$.

(f) We observe that setting $a = -1/2$ in (c) gives

$$(1-x^2)^{-1/2} = \sum_{k=0}^{\infty} \binom{-1/2}{k} (-x^2)^k = \sum_{k=0}^{\infty} \frac{(2k-1)!!}{(2k)!!} x^{2k},$$

because

$$\binom{-1/2}{k} = \left(-\frac{1}{2}\right) \left(-\frac{1}{2}-1\right) \cdots \left(-\frac{1}{2}-k+1\right) / k! = (-1)^k \frac{(2k-1)!!}{(2k)!!}.$$

(Note that $(2k)!! = 2^k k!$.) Termwise integration according to the corollary following Theorem 5.3 yields

$$\arcsin x = \sum_{k=0}^{\infty} \frac{(2k-1)!!}{(2k)!!(2k+1)} x^{2k+1} + C.$$

In order to get equality when $x = 0$, we must have $C = 0$. ■

Remark. Observe two special cases of (c). When $a = n$ is a positive integer, we recover the ordinary binomial theorem, and when $a = -1$, we get the geometric series $\sum_0^{\infty} (-x)^k$.

Power Series Solutions of Differential Equations

In the proof of Theorem 5.4 we determined the sum $s(x)$ of various power series by finding differential equations for s that we were able to solve. Often the situation is the reversed: We have an ordinary differential equation that cannot be solved in the usual way and therefore we seek a solution in form of a power series. We give some examples.

Example 5.6. Find in the form of a power series $\sum_{k=0}^{\infty} a_k x^k$ a solution of the differential equation

$$4xu'' + 2u' - u = 0, \quad x > 0,$$

with the initial condition $u(0) = 1$.

Solution: Assume that there is a power series solution $u(x)$ with positive radius of convergence R . Then

$$u'(x) = \sum_{k=1}^{\infty} a_k k x^{k-1}, \quad u''(x) = \sum_{k=2}^{\infty} a_k k(k-1) x^{k-2} \quad \text{when} \quad |x| < R.$$

We substitute this in the equation and get

$$\sum_{k=2}^{\infty} 4a_k k(k-1) x^{k-1} + \sum_{k=1}^{\infty} 2a_k k x^{k-1} - \sum_{k=0}^{\infty} a_k x^k = 0.$$

In the first series, we can start the summation with $k = 1$ since the corresponding term is zero. Replacing the index k with $k - 1$ in the last series and adding the sums, we get

$$\sum_{k=1}^{\infty} (4a_k k(k-1) + 2a_k k - a_{k-1}) x^{k-1} = 0.$$

This is a power series the sum of which is zero. Hence all the coefficients must be zero since the coefficients are uniquely determined by the sum. Therefore,

$$4a_k k(k-1) + 2a_k k - a_{k-1} = 0, \quad k = 1, 2, \dots,$$

and this is equivalent to

$$a_k = \frac{1}{2k(2k-1)} a_{k-1}, \quad k = 1, 2, \dots \quad (19)$$

Since $u(0) = 1$, $a_0 = 1$. Hence using the recurrence formula (19), we obtain successively

$$a_1 = \frac{1}{2}, \quad a_2 = \frac{1}{4 \cdot 3 \cdot 2} = \frac{1}{24}, \quad a_3 = \frac{1}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2} = \frac{1}{720},$$

and, in general,

$$a_k = \frac{1}{(2k)!}, \quad k = 1, 2, \dots$$

Therefore

$$u(x) = \sum_{k=0}^{\infty} \frac{1}{(2k)!} x^k. \quad (20)$$

Note that the preceding calculations were formal. We assumed that there exists a power series solution of the differential equation. However, using d'Alembert's ratio test, we see that the power series (20) has radius of convergence $R = \infty$. This is so because it follows from (19) that

$$\frac{a_k}{a_{k-1}} = \frac{1}{2k(2k-1)} \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

This justifies the calculations and we have found the solution

$$u(x) = \sum_{k=0}^{\infty} \frac{1}{(2k)!} x^k$$

of the differential equation.

Remark. In this case, the solution $u(x)$ can be expressed in elementary functions. We have that $u(x) = \frac{1}{2}(e^{\sqrt{x}} + e^{-\sqrt{x}}) = \cosh \sqrt{x}$ when $x \geq 0$. In the general case, this is not possible and one has to settle for a solution in the form of a power series.

Example 5.7. Consider Airy's differential equation

$$u'' - xu = 0.$$

Setting $u(x) = \sum_0^\infty a_k x^k$, we get, as in the preceding example, that

$$\sum_{k=2}^\infty a_k k(k-1)x^{k-2} - \sum_{k=0}^\infty a_k x^{k+1} = 0.$$

We replace k with $k+3$ in the first series. The summation then starts with $k = -1$. Writing the first term of the first series separately and collecting all the other terms into a single sum, we get the equation

$$2a_2 + \sum_{k=0}^\infty (a_{k+3}(k+3)(k+2) - a_k)x^{k+1} = 0.$$

It follows that

$$a_2 = 0 \quad \text{and} \quad (k+3)(k+2)a_{k+3} - a_k = 0, \quad k = 0, 1, 2, \dots$$

Hence all the coefficients a_k are uniquely determined by the values of a_0 and a_1 . To be able to write the formulae conveniently, we introduce some notation. Set

$$q_n = \frac{1}{3n(3n-1)}, \quad r_n = \frac{1}{(3n+1)3n}, \quad n = 1, 2, \dots,$$

and

$$Q_n = q_n q_{n-1} \cdots q_1, \quad R_n = r_n r_{n-1} \cdots r_1.$$

Then the recurrence formula yields

$$\begin{aligned} a_{3n} &= a_{3n-3} q_n = a_{3n-6} q_n q_{n-1} = \cdots = a_0 Q_n, \\ a_{3n+1} &= a_{3n-2} r_n = a_{3n-5} r_n r_{n-1} = \cdots = a_1 R_n, \\ a_{3n+2} &= \cdots = 0, \end{aligned}$$

where we used that $a_2 = 0$ in the last line. If we set $Q_0 = R_0 = 1$, then these formulae hold also when $n = 0$. Since $a_0 = u(0)$ and $a_1 = u'(0)$, we get that

$$\begin{aligned} u(x) &= a_0 \sum_{n=0}^\infty Q_n x^{3n} + a_1 \sum_{n=0}^\infty R_n x^{3n+1} \\ &= u(0) \sum_{n=0}^\infty Q_n x^{3n} + u'(0)x \sum_{n=0}^\infty R_n x^{3n}. \end{aligned}$$

In order to show that this is a solution, we must verify that the radii of convergence of the power series are positive. In fact, it follows from d'Alembert's ratio test that the series are convergent for all x because

$$\frac{Q_{n+1}}{Q_n} = q_{n+1} \rightarrow 0 \quad \text{as} \quad n \rightarrow \infty \quad \text{and} \quad \frac{R_{n+1}}{R_n} = r_{n+1} \rightarrow 0 \quad \text{as} \quad n \rightarrow \infty.$$

Therefore, there exists a power series $u(x)$ that is a solution of Airy's differential equation and has arbitrarily prescribed values of $u(0)$ and $u'(0)$. (It can be shown that there is only one such solution and hence it is the one we found here.)

Finally, we would like to point out that not all differential equations have power series solutions. For instance, consider the equation

$$2xu' - u = 0, \quad x > 0.$$

It is plainly seen that the solutions of this equation are $u(x) = C\sqrt{x}$ where C is a constant. Since the function $\sqrt{x}$ is not differentiable at the origin, it cannot be the sum of a convergent power series. (The reader is asked to examine what the consequences are of the assumption of a power series solution of this differential equation.)

Exercises for 1.5 Power Series

115. Determine the radii of convergence of the following power series.

- | | |
|---|--|
| (a) $\sum_0^{\infty} k^2 x^k,$ | (b) $\sum_0^{\infty} k^k x^k,$ |
| (c) $\sum_0^{\infty} \frac{x^k}{(k!)^2},$ | (d) $\sum_0^{\infty} 2^{-k} x^k,$ |
| (e) $\sum_2^{\infty} \frac{x^k}{\ln k},$ | (f) $\sum_1^{\infty} \frac{2^k}{k^2} x^k,$ |
| (g) $\sum_1^{\infty} \frac{x^k}{\arctan x},$ | (h) $\sum_0^{\infty} (k^2 + 2^k) x^k,$ |
| (i) $\sum_0^{\infty} \left(\cos \frac{k\pi}{4} \right) x^k,$ | (j) $\sum_1^{\infty} \frac{2^k + 2^{-k}}{k} x^k,$ |
| (k) $\sum_1^{\infty} k^{1/k} x^k,$ | (l) $\sum_0^{\infty} (2^k + (-2)^k + 1) x^k,$ |
| (m) $\sum_1^{\infty} \left(\sqrt{k^2 + 1} - \sqrt{k^2 - 1} \right) x^k,$ | (n) $\sum_0^{\infty} \frac{(k+2)^3}{3^{2k+1}} x^k,$ |
| (o) $\sum_1^{\infty} \frac{2^{1/k}}{2^k} x^k,$ | (p) $\sum_1^{\infty} \frac{2^k}{\sqrt{k}} x^{2k},$ |
| (q) $\sum_1^{\infty} \frac{x^k}{1 + 2 + \cdots + k},$ | (r) $\sum_1^{\infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{k} \right) x^k,$ |
| (s) $\sum_0^{\infty} \frac{x^{4k}}{2^k + k^2},$ | (t) $\sum_0^{\infty} (\sin k) x^k.$ |

116. Compute the radius of convergence of $\sum_1^{\infty} a_k x^k$ when a_k equals

- | | |
|--|---|
| (a) $\frac{k^k}{k!},$ | (b) $\arctan \frac{1}{k},$ |
| (c) $\binom{a}{k} = \frac{a(a-1)\cdots(a-k+1)}{k!}, \quad a \in \mathbf{R},$ | (d) $\frac{(3k+1)!}{(k!)^4},$ |
| (e) $2^{\sqrt{k}},$ | (f) $\left(1 + \frac{1}{k} \right)^{k^2}.$ |

117. Compute the sums of the following series when $|x| < 1$.

- | | | |
|--------------------------------------|------------------------------------|---------------------------------------|
| (a) $\sum_{k=1}^{\infty} k x^{k-1},$ | (b) $\sum_{k=1}^{\infty} k x^k,$ | (c) $\sum_{k=2}^{\infty} k(k-1) x^k,$ |
| (d) $\sum_{k=1}^{\infty} k^2 x^k,$ | (e) $\sum_{k=1}^{\infty} k^3 x^k.$ | |

118. Compute

$$(a) \sum_{k=0}^{\infty} \frac{x^k}{k+1}, \quad |x| < 1, \quad (b) \sum_{k=0}^{\infty} \frac{x^k}{k!(k+2)}, \quad x \in \mathbf{R}.$$

119. Show that

$$\sum_{k=1}^{\infty} \frac{3^k + 3^{-k}}{k} x^k = \ln \frac{3}{(1-3x)(3-x)}, \quad |x| < \frac{1}{3}.$$

120. Determine in each of the following five cases whether or not there exist numbers a_k , $k = 0, 1, \dots$, and a number $R > 0$ such that

$$(a) \sum_{k=0}^{\infty} a_k x^k = \frac{x}{1+x^2} \quad \text{when} \quad -R < x < R,$$

$$(b) \sum_{k=0}^{\infty} a_k x^k = \begin{cases} 1 & \text{when } 0 < x < R, \\ -1 & \text{when } -R < x < 0, \end{cases}$$

$$(c) \sum_{k=0}^{\infty} a_k x^k = |x|, \quad -R < x < R,$$

$$(d) \sum_{k=0}^{\infty} a_k x^k = \begin{cases} x^2 & \text{when } 0 < x < R, \\ -x^2 & \text{when } -R < x < 0, \end{cases}$$

$$(e) \sum_{k=0}^{\infty} a_k x^k = \begin{cases} \cosh \sqrt{x} & \text{when } 0 < x < R, \\ \cos \sqrt{-x} & \text{when } -R < x < 0, \end{cases}$$

where

$$\cosh t = \frac{e^t + e^{-t}}{2}.$$

121. Use the Taylor series of the elementary functions to show that

$$(a) a^x = \sum_{k=0}^{\infty} \frac{(\ln a)^k}{k!} x^k, \quad x \in \mathbf{R}, \quad a > 0,$$

$$(b) \cosh x = \frac{e^x + e^{-x}}{2} = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}, \quad x \in \mathbf{R},$$

$$(c) \sinh x = \frac{e^x - e^{-x}}{2} = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}, \quad x \in \mathbf{R},$$

$$(d) \ln \sqrt{\frac{1+x}{1-x}} = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{2k+1}, \quad |x| < 1,$$

$$(e) 2 \sin^2 \frac{x}{2} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{(2k)!} x^{2k}, \quad x \in \mathbf{R},$$

$$(f) (1-x) \ln(1-x) = -x + \sum_{k=2}^{\infty} \frac{x^k}{(k-1)!}, \quad |x| < 1,$$

$$(g) \sqrt{\frac{1+x}{1-x}} = \sum_{k=0}^{\infty} \frac{(2k-1)!!}{(2k)!!} (x^{2k} + x^{2k+1}), \quad |x| < 1.$$

122. Solve the differential equation

$$(1 - x^2) u''(x) - 2xu'(x) + 20u(x) = 0, \quad u(0) = 1, \quad u'(0) = 0,$$

by seeking a power series solution $u(x) = \sum_0^\infty a_k x^k$.

123. Find a power series solution of

$$(a) \quad x^2 u''(x) + 3xu'(x) + u(x) = \frac{1}{1-x}, \quad |x| < 1,$$

$$(b) \quad x^2 u''(x) + \frac{1}{4}u(x) = \ln(1+x), \quad |x| < 1.$$

124. Find all power series solutions of

$$xu''(x) = u(x), \quad x \in \mathbf{R}.$$

125. Assume that the problem

$$xu''(x) + u'(x) - u(x) = 0, \quad u(0) = 1,$$

has a power series solution and show that this leads to the conclusion that

$$u(x) = \sum_{k=0}^{\infty} \frac{x^k}{(k!)^2}.$$

126. Solve the problem

$$x^2 u''(x) + (1 + x^2)u'(x) + u(x) = 0, \quad u(0) = 1,$$

by seeking a power series solution. The answer should be given in elementary functions.

127. Examine whether or not the differential equation

$$x^2 u''(x) - u'(x) + u(x) = 0$$

has a power series solution with positive radius of convergence.

Answers to Exercises for 1.5 Power Series

115. (a) 1, (b) 0, (c) ∞ , (d) 2, (e) 1,
 (f) $\frac{1}{2}$, (g) 1, (h) $\frac{1}{2}$, (i) 1, (j) $\frac{1}{2}$,
 (k) 1, (l) $\frac{1}{2}$, (m) 1, (n) 9, (o) 2,
 (p) $\frac{1}{\sqrt{2}}$, (q) 1, (r) 1, (s) $2^{1/4}$, (t) 1.
116. (a) $\frac{1}{e}$, (b) 1, (c) ∞ when $a \in \mathbf{N}$, 1 when $a \in \mathbf{R} \setminus \mathbf{N}$,
 (d) ∞ , (e) 1, (f) $\frac{1}{e}$.
117. (a) $\frac{d}{dx}(1-x)^{-1} = (1-x)^{-2}$, (b) $x(1-x)^{-2}$,
 (c) $x^2 \frac{d^2}{dx^2}(1-x)^{-1} = 2x^2(1-x)^{-3}$, (d) $\frac{x+x^2}{(1-x)^3}$,
 (e) $\frac{x+4x^2+x^3}{(1-x)^4}$.
118. (a) $-\frac{\ln(1-x)}{x}$ when $0 < |x| < 1$, 1 when $x = 0$,
 (b) $\frac{xe^x - e^x + 1}{x^2}$ when $x \neq 0$, $\frac{1}{2}$ when $x = 0$.
120. (a) Yes, (b) no, (c) no, (d) no, (e) yes.
122. $u(x) = 1 - 10x^2 + \frac{35}{3}x^4$.
123. (a) $\sum_{k=0}^{\infty} \frac{1}{(k+1)^2} x^k$, (b) $\sum_{k=1}^{\infty} \frac{4(-1)^{k-1}}{k(2k-1)^2} x^k$.
124. $\sum_0^{\infty} a_k x^k$ where

$$a_k = \frac{k}{(k!)^2} a_1$$
 and a_1 is an arbitrary real constant.
126. $u(x) = e^{-x}$.
127. No, because $(k+1)a_{k+1} = (k^2 - k + 1)a_k$ and hence (why?) $R = 0$ if $a_0 \neq 0$.